All-cuts CODI epoch40 | validation_equations | common-m-ge6
Same samples across every column | population-weighted token CE | lower is better

Full suffix + answer CE

| Latent cut k | 1/12 | 1/6 | 1/4 | 1/3 | 5/12 | 1/2 | 7/12 | 2/3 | 3/4 | 5/6 | 11/12 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 (control) | 0.79 | 0.81 | 0.90 | 0.99 | 1.07 | 1.15 | 1.45 | 1.45 | 2.09 | 2.09 | 2.09 |
| 1 | 0.49 | 0.48 | 0.55 | 0.62 | 0.65 | 0.67 | 0.76 | 0.76 | 0.93 | 0.93 | 0.93 |
| 2 | 0.49 | 0.48 | 0.55 | 0.62 | 0.65 | 0.67 | 0.77 | 0.77 | 0.94 | 0.94 | 0.94 |
| 3 | 0.50 | 0.48 | 0.53 | 0.59 | 0.62 | 0.64 | 0.76 | 0.76 | 0.93 | 0.93 | 0.93 |
| 4 | 0.50 | 0.49 | 0.54 | 0.60 | 0.61 | 0.64 | 0.76 | 0.76 | 0.94 | 0.94 | 0.94 |
| 5 | 0.50 | 0.48 | 0.54 | 0.60 | 0.60 | 0.64 | 0.76 | 0.76 | 0.93 | 0.93 | 0.93 |
| 6 | 0.56 | 0.55 | 0.61 | 0.68 | 0.69 | 0.73 | 0.90 | 0.90 | 1.15 | 1.15 | 1.15 |

nats / token: 0.6, 0.8, 1.0, 1.2, 1.4, 1.6, 1.8, 2.0

First-step content CE (exclude <<)

| Latent cut k | 1/12 | 1/6 | 1/4 | 1/3 | 5/12 | 1/2 | 7/12 | 2/3 | 3/4 | 5/6 | 11/12 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 (control) | 0.90 | 0.91 | 0.90 | 0.90 | 1.02 | 1.00 | 0.90 | 0.90 | 1.28 | 1.28 | 1.28 |
| 1 | 0.84 | 0.75 | 0.83 | 0.92 | 1.05 | 1.01 | 0.92 | 0.92 | 1.20 | 1.20 | 1.20 |
| 2 | 0.84 | 0.74 | 0.83 | 0.91 | 1.08 | 1.04 | 0.94 | 0.94 | 1.20 | 1.20 | 1.20 |
| 3 | 0.86 | 0.75 | 0.78 | 0.83 | 0.96 | 0.95 | 0.94 | 0.94 | 1.20 | 1.20 | 1.20 |
| 4 | 0.86 | 0.75 | 0.79 | 0.85 | 0.96 | 0.94 | 0.93 | 0.93 | 1.19 | 1.19 | 1.19 |
| 5 | 0.87 | 0.76 | 0.80 | 0.84 | 0.95 | 0.95 | 0.95 | 0.95 | 1.19 | 1.19 | 1.19 |
| 6 | 0.87 | 0.75 | 0.81 | 0.86 | 0.95 | 0.95 | 0.95 | 0.95 | 1.19 | 1.19 | 1.19 |

nats / token: 0.8, 0.9, 1.0, 1.1, 1.2

Explicit cut ratio r (ceil + clamp)

Full suffix + answer CE

| Latent cut k | 1 n=8 | 2 n=8 | 3 n=8 | 4 n=8 | 5 n=8 |
|---|---|---|---|---|---|
| 0 (control) | 0.79 | 0.90 | 1.07 | 1.30 | 1.85 |
| 1 | 0.49 | 0.55 | 0.65 | 0.71 | 0.94 |
| 2 | 0.49 | 0.55 | 0.65 | 0.72 | 0.94 |
| 3 | 0.50 | 0.53 | 0.62 | 0.70 | 0.93 |
| 4 | 0.50 | 0.54 | 0.61 | 0.71 | 0.94 |
| 5 | 0.50 | 0.54 | 0.60 | 0.70 | 0.93 |
| 6 | 0.56 | 0.61 | 0.69 | 0.82 | 1.10 |

Explicit prefix length j (predict from step j+1)

nats / token: 0.6, 0.8, 1.0, 1.2, 1.4, 1.6, 1.8

Explicit cut ratio r (ceil + clamp)

First-step content CE (exclude <<)

| Latent cut k | 1 n=8 | 2 n=8 | 3 n=8 | 4 n=8 | 5 n=8 |
|---|---|---|---|---|---|
| 0 (control) | 0.90 | 0.90 | 1.02 | 0.71 | 1.20 |
| 1 | 0.84 | 0.83 | 1.05 | 0.72 | 1.18 |
| 2 | 0.84 | 0.83 | 1.08 | 0.76 | 1.17 |
| 3 | 0.86 | 0.78 | 0.96 | 0.74 | 1.19 |
| 4 | 0.86 | 0.79 | 0.96 | 0.74 | 1.18 |
| 5 | 0.87 | 0.80 | 0.95 | 0.75 | 1.18 |
| 6 | 0.87 | 0.81 | 0.95 | 0.77 | 1.18 |

Explicit prefix length j (predict from step j+1)

nats / token: 0.8, 0.9, 1.0, 1.1